

CS & IT ENGINEERING



'C' PROGRAMMING

Practice Session

Lecture No.- 02



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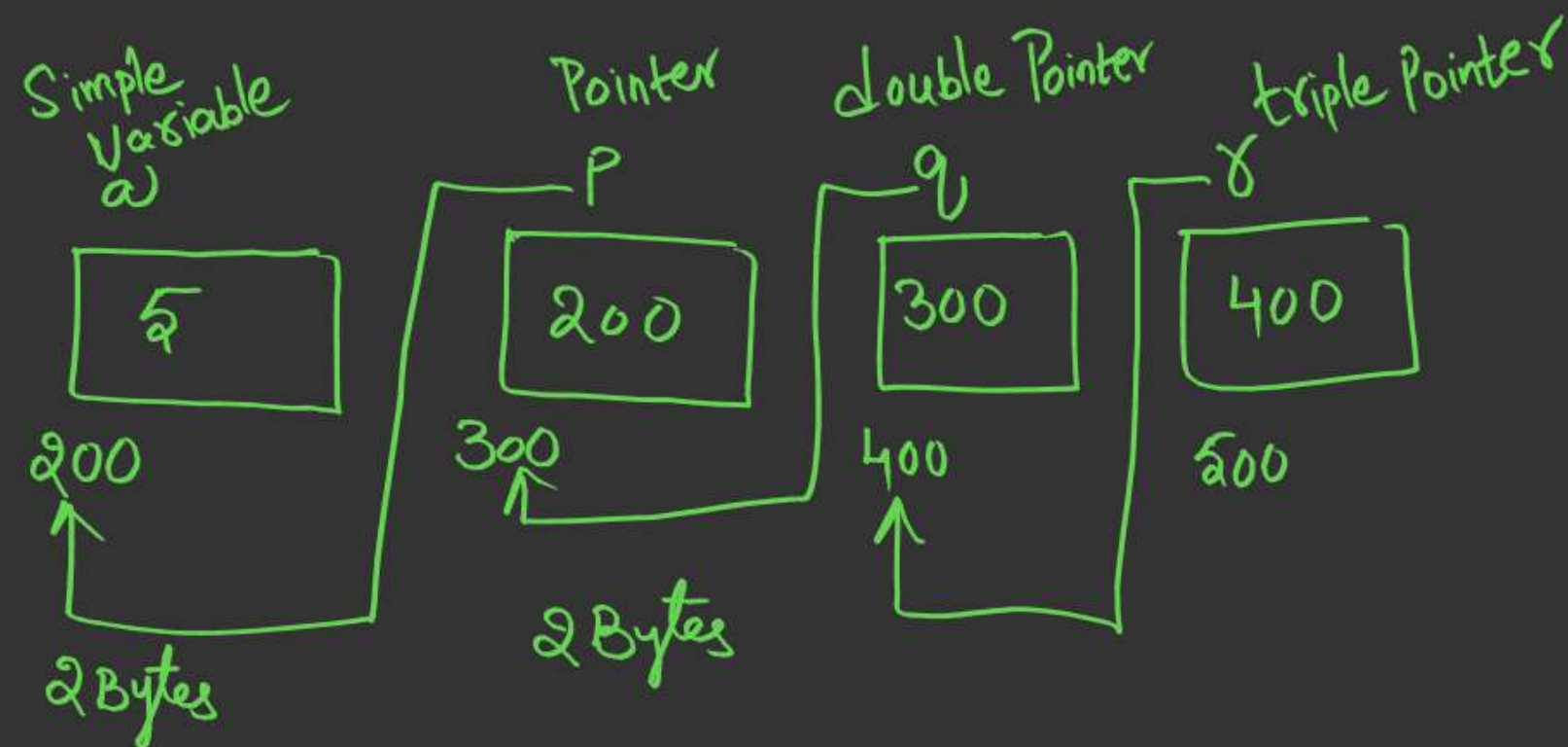
Topics to be Covered



- Practice session

Pointer to Pointer / Pointer Using Pointer

```
int a=5, *p=&a, **q=&p, ***x=&q;
```



```
Printf("%d", a); // 5
```

```
Printf("%d", *p); // 5
```

```
Printf("%d", **q); // 5
```

```
Printf("%d", ***x); // 5
```

→ Indirection operator
(or)
dereference operator

3-D array

`int x[2][3][4] = {1, 2, 3, 5, 7, 9, 11, 13, 15, 21, 23, 25, 27, 29, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22}`

No. of 2-D arrays

3 1-D arrays
in each 2-D array

4 Elements in Each
1-D array

1st index = 0 3rd index

2nd index	0	1	2	3
0	1	2	3	5
1	7	9	11	13
2	15	21	23	25

`x[0][2][1]`

1st index = 1 3rd index

2nd index	0	1	2	3
0	27	29	4	6
1	8	10	12	14
2	16	18	20	22

`x[0][0][3]`

`x[1][0][1]`

`x[1][1][3]`



Topic : Practice Session



#Q. Consider the following C program :

```
#include <stdio.h>
```

```
int main () {
```

```
    int arr [] = {1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip = arr+4;
```

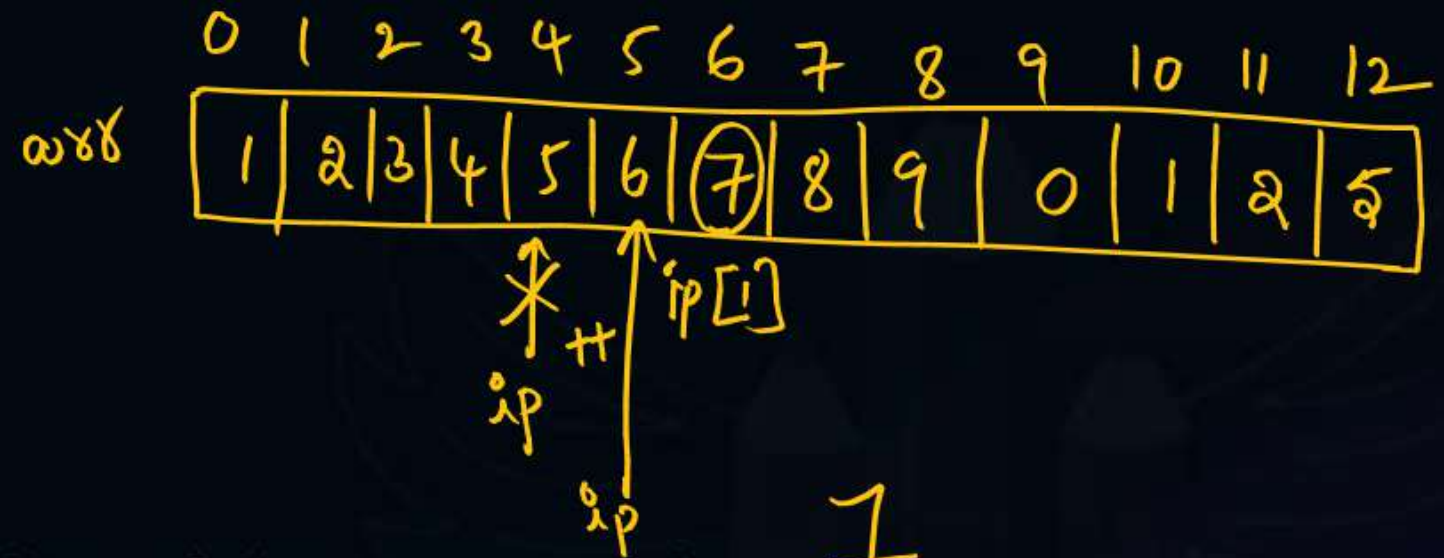
```
    ip= ip++;
```

```
    printf ("%d\n", ip[1]);
```

```
    return 0;
```

```
}
```

→ $*(ip+1)$



The number that will be displayed on execution of the program is 7.



#Q. Consider the following C program segment.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

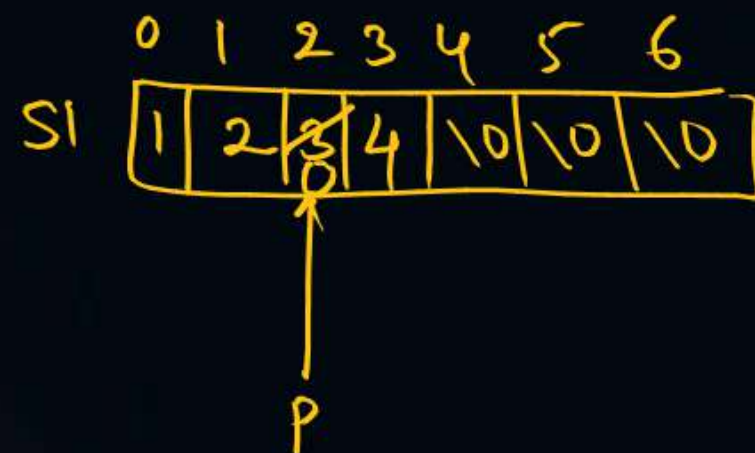
```
char s1[7] = "1234", *p;
```

```
p = s1 + 2;
```

```
*p = '0';
```

```
printf("%s", s1);
```

```
}
```



What will be printed by the program?

A. 12

B. 120400

☒ C. 1204

D. 1034



#Q. Consider the following program.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int arr[4][5];
```

```
int i, j;
```

```
for (i=0; i<4; i++) {
```

```
for (j=0; j<5; j++) {
```

```
arr[i][j] = 10 * i + j;
```

```
}
```

```
}
```

```
printf("%d", *(arr[1]+9));
```

```
return 0;
```

```
}
```

What is the output of the above program?

A. 14

B. 20

✓ C. 24

D. 30

arr	0	1	2	3	4
0	0	1	2	3	4
1	10	11	12	13	14
2	20	21	22	23	24
3	30	31	32	33	34

$$110 + 9 * 2 = 110 + 18 = 128$$

$$\rightarrow arr[1] + 9$$

$$\rightarrow *(arr[1] + 9)$$

$$* (128) = 24$$



#Q. What is the output printed by the following C code?

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
char a [6] = "world";
```

```
int i, j;
```

```
for (i = 0, j = 5; i < j; a [i++] = a [j--]);
```

```
printf ("%s\n", a);
```

```
}
```

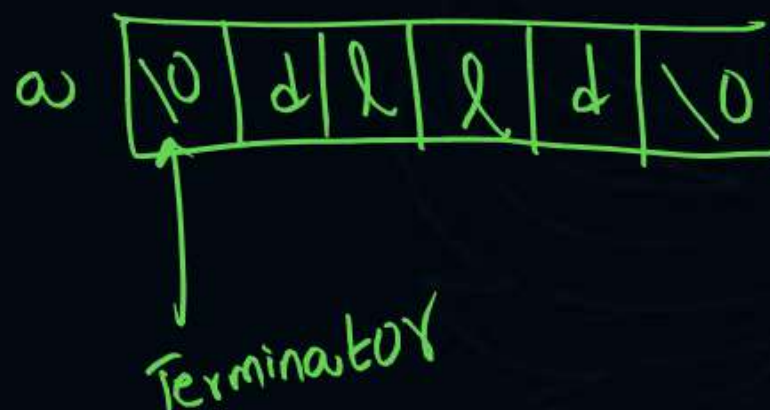
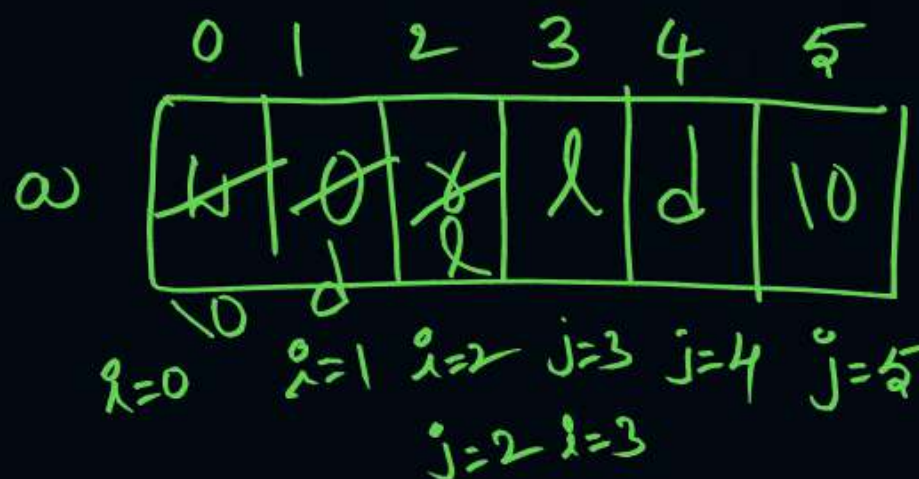
$a[i] = a[j]$
 $i++, j--$

A. dlrow

☒ B. Null string (Nothing Printed)

C. dlrd

D. worow





#Q. Consider the following C program:

```
#include <stdio.h>
int main( )
{
    int a[4][5] = { {1, 2, 3, 4, 5},
                    {6, 7, 8, 9, 10},
                    {11, 12, 13, 14, 15},
                    {16, 17, 18, 19, 20} };

```

```
    printf("%d"\n, *(*a+**a+1)+4));
```

```
    return(0);
```

```
}
```

The output of the program is 15. $*(*a+**a+1) = a[0][0] = 1$ $(a+1+1) = (a+2)$
 $*(*a+2) = a[0][2] = 3$ $(3+4) = 7$

a	0	1	2	3	4
0	1	2	3	4	5
1	6	7	8	9	10
2	11	12	13	14	15
3	16	17	18	19	20



#Q. Consider the C program given below. What does it print?

```
#include <stdio.h>
```

```
int main () {
```

```
int i, j;
```

```
int a [8] = {1, 2, 3, 4, 5, 6, 7, 8};
```

```
for(i = 0; i < 3; i++) {
```

```
    a[i] = a[i] + 1;
```

```
    i++;
```

//End of for loop

```
    i--; i = 3
```

```
    for (j = 7; j > 4; j--) {
```

```
        int i = j/2;
```

```
        a[i] = a[i] - 1; } //End of for
```

```
        printf ("%d, %d", i, a[i]);
```

```
    }
```

ω

0	1	2	3	4	5	6	7
1	2	3	4	5	6	7	8

2 4 3
3 2

i = 0 a[0] = a[0] + 1, i = 1

i = 2 a[2] = a[2] + 1, i = 3,

i = 4 4 < 3 False

j = 7 i = 3
a[3] = a[3] - 1

j = 6
i = 3
a[3] = a[3] - 1

j = 5
i = 5/2 = 2
a[2] = a[2] - 1

j = 4
4 > 4 False.

A. 2, 3

B. 2, 4

☒ C. 3, 2

D. 3, 3

3, 2

3, 2



Topic : Practice Session – GATE 2008



#Q. C program is given below:

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
int i, j;
```

```
char a [2] [3] = {{ 'a', 'b', 'c' }, { 'd', 'e', 'f' }};
```

```
char b [3] [2];
```

```
char *p = *b;
```

```
for (i = 0; i < 2; i++) {
```

```
for (j = 0; j < 3; j++) {
```

```
*(p + 2*j + i) = a[i][j];
```

```
} i=0 j=0 * (P+0+0) = a[0][0]
```

```
} * (P+0) = a[0][0]
```

```
}
```

What should be the contents of the array b at the end of the program?

		b		
		0	1	
P → 0	a	P+0	d	P+1
1	b	P+2	e	P+3
2	c	P+4	f	P+5

A.

a b

c d

e f

B.

a d

b e

c f

C.

a c

e b

d f

D.

a e

d c

b f

		0	1	2
		0	1	2
0	a	b	c	
1	d	e	f	

i=1 j=0 * (P+1) = a[1][0]

j=1 * (P+3) = a[1][1]

j=2 * (P+5) = a[1][2]

j=2

* (P+4+0) = a[0][2]

* (P+4) = a[0][2]

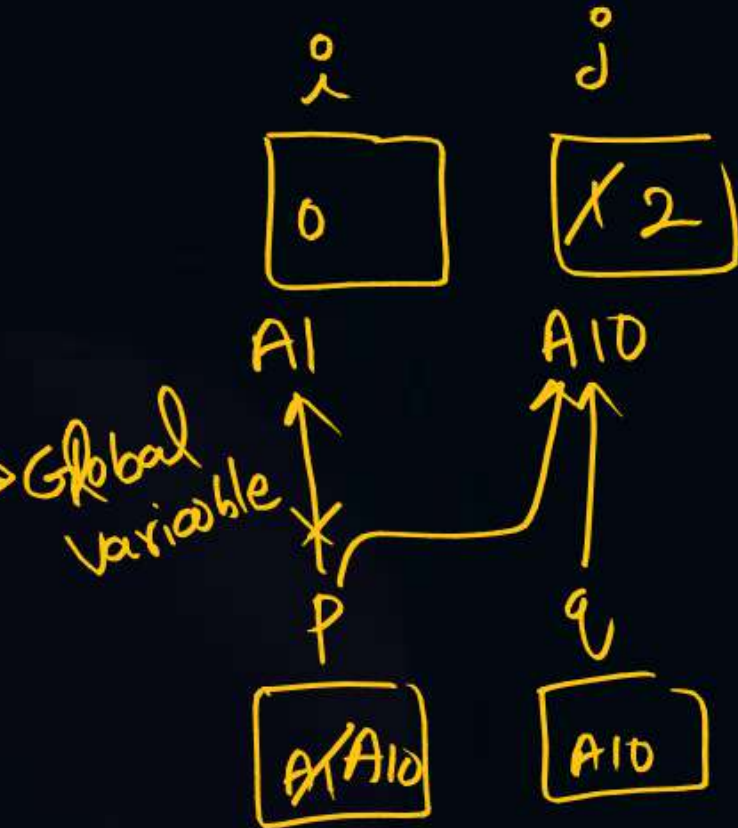


Topic : Practice Session



#Q. What does the following program print?

```
void f(int *p, int *q)
{
    p = q;
    *p = 2;
}
int i = 0, j = 1;
int main()
{
    f(&i, &j);
    printf("%d %d \n", i, j);
    getchar();
    return 0;
}
```



a) 2 2

b) 2 1

c) 0 1

d) 0 2



#Q. What is printed by the following C program?

```
int f(int x, int *py, int **ppz)
```

```
{
```

```
int y, z;
```

```
**ppz += 1; z = **ppz;
```

```
*py += 2; y = *py;
```

```
x += 3;
```

```
return x+y+z;
```

```
}
```

```
void main()
```

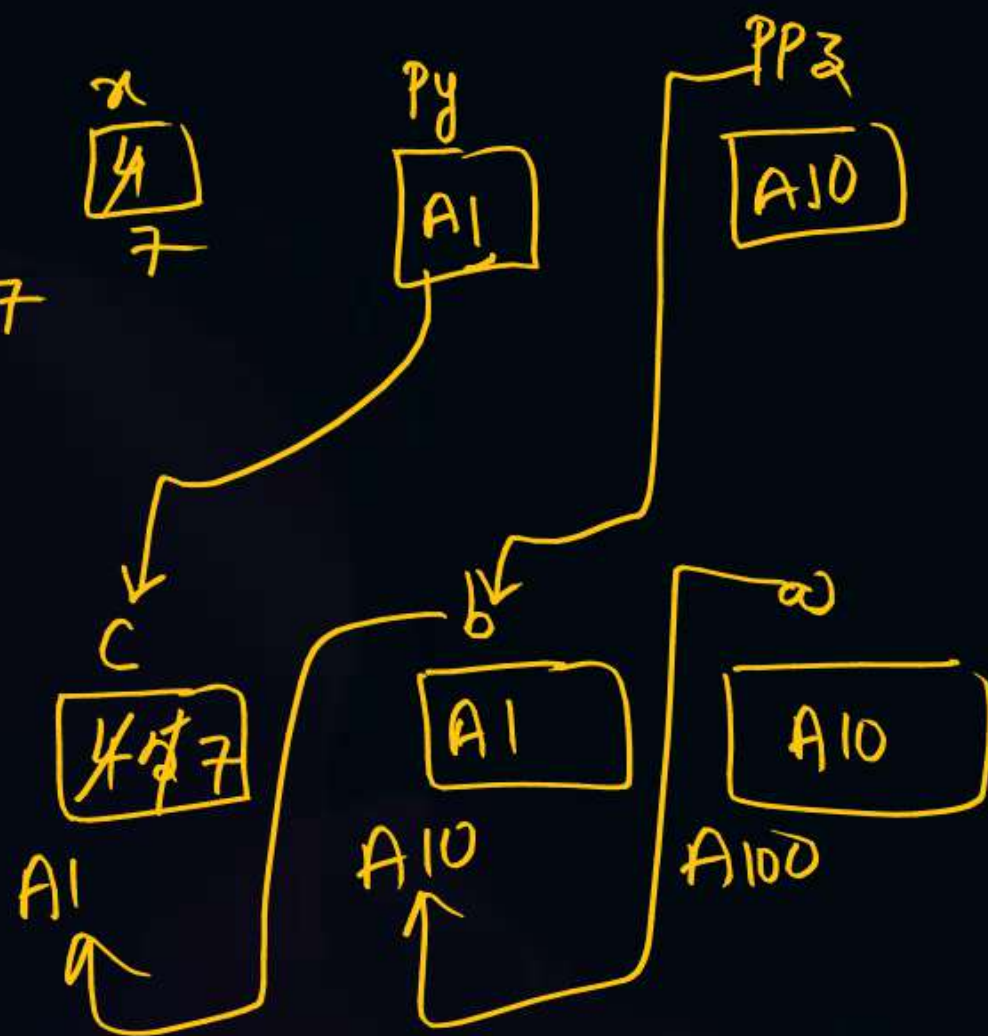
```
{
```

```
int c, *b, **a;
```

```
c = 4; b = &c; a = &b;
```

```
printf("%d", f(c, b, a));
```

```
}
```



A. 18

☒ B. 19

C. 20

D. 21

o/p: 19

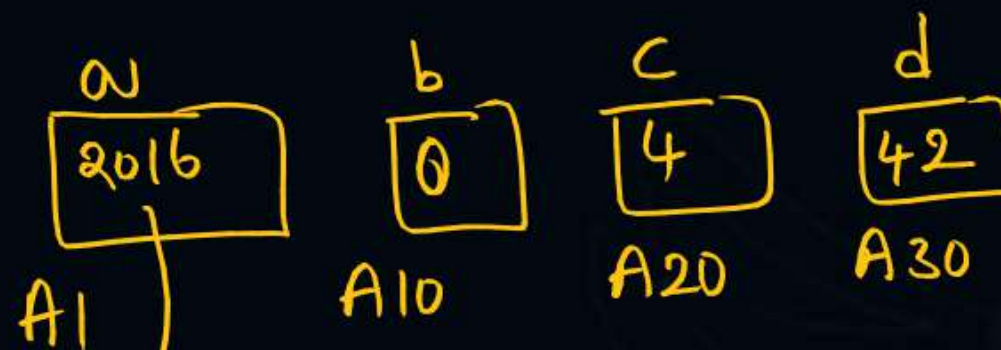


#Q. Consider the following C program.

```
#include <stdio.h>
void mystery (int *ptrA1a, int *ptrA30b) {
int *temp;
temp = ptrb;
ptrb = ptrA1a;
ptrA10a = temp;
}
```

```
→ int main () {
int a = 2016, b=0, c= 4, d = 42;
mystery (&a, &b);
if (a < c) 2016 < 4 False
mystery (&c, &a); X
mystery (&a, &d);
printf ("%d\n", a);
}
```

The output of the program is 2016.





H/W

```
#Q. int pp(int a, int b) {  
    int arr[20];  
    int i, tot = 1, ex, len;  
    ex = a;  
    len = tob (b, arr);  
    for (i=0; i < len; i++) {  
        if (arr[i] ==1)  
            tot = tot * ex;  
        ex = ex * ex;  
    }  
    return (tot);  
}
```

Consider the following C functions.

```
int tob(int b, int* arr) {  
    int i;  
    for (i=0; b>0; i++) {  
        if (b%2) arr[i] = 1;  
        else arr[i]=0;  
        b = b/2;  
    }  
    return (i);  
}
```

The value returned by pp (3, 4) is _____.



THANK - YOU